

Determinant of 2x2 matrices

Key Points

The determinant of a 2x2 matrix M is denoted by $|M|$ and the formula is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

That is the product of the leading diagonal minus the product of the trailing diagonal

It represents the area scale factor of a matrix transform

Alternative notation is $\det(M)$

Basic Skills

Q1. Find the following determinants

a) $\begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix}$ b) $\begin{vmatrix} -3 & -1 \\ 2 & 1 \end{vmatrix}$ c) $\begin{vmatrix} 4 & 2 \\ -2 & 4 \end{vmatrix}$

Q2. The area of a shape before a transform is 8, what is the area after the transform by $M = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$

Q3. Explain geometrically why the matrix transform of a reflection in $y = x$ has a determinant of 1

Q4. A matrix transform $M = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$ preserves the area of any shape, what is the relationship between a, d

Competency Skills

Q5. Given that the determinant of matrix $M = \begin{pmatrix} x & -3 \\ 3 & x-5 \end{pmatrix}$ is 5 find x

Q6. Given that $\begin{vmatrix} x & y \\ 5 & 1 \end{vmatrix} = 2$ and $\begin{vmatrix} x & 2 \\ y & 3 \end{vmatrix} = 5$ find x, y

Q7. The 2D Euclidean plane is transformed by $M = \begin{pmatrix} 5 & 4 \\ 2 & 3 \end{pmatrix}$

- a) Show on a diagram the transform of the unit square
- b) Describe the effect of the transform with reference to the determinant

Q8. Prove that $|kM| = k^2|M|$ for a 2x2 matrix

Q9. For the general 2x2 matrix transform $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ work out the area of the transformed unit square by subtracting 2 trapeziums and 2 triangles from a rectangle

Advanced Skills

Q10. Given M maps all points on $y = x + 2$ to the point (1,2)

- a) Find M and show the determinant is zero
- b) Show that all points are mapped to the line $y = 2x$
- c) Find the line which is transformed to the point (0,0)

Q11. Determine if the following properties for the determinant are true, proving or disproving them

- a) $\det(AB) = \det(A) \det(B)$
- b) $\det(A + B) = \det(A) + \det(B)$

Extension

Q12. What does a negative determinant mean in terms of the transformed area